

NUMBER SYSTEM CONCEPTS NOTES

Integers + terminating + non-terminating & repeating

CLASSIFICATIONS OF NUMBERS :-

Numbers:
 (A) Real No's. $\rightarrow$ any no. on No. line
 (B) Imaginary No's $\rightarrow$ not on No. line
 (C) Complex No's. $\rightarrow$ real + img.

$\rightarrow$ (1) is neither prime nor composite (1 is inverse of 1)

PPNNO :-

B/W.

b/w 1-100 $\rightarrow$ 25

b/w 1-50 $\rightarrow$ 15

b/w 50-100 $\rightarrow$ 10

b/w 1-200 $\rightarrow$ 46

b/w 1-500 $\rightarrow$ 95

b/w 1-1000 $\rightarrow$ 168

b/w 1-100 Nos $\rightarrow$ smallest two digit P.No
 2, 3, 5, 7, (11), 13, 17, 19, 23, 29, 31, 37,
 41, 43, 47, 53, 59, 61, 67, 71, 73, 79,
 83, 89, (97)
 largest 2 digit P.No.

Twin P.No's :- whose diff. is 2.

eg $\rightarrow$ (3, 5), (17, 19), (11, 13) -

Co-prime (relatively prime) :- whose HCF = 1
 eg $\rightarrow$ (15, 23), (7, 13), (9, 14), (11, 16),
 (9, 26).

Integers :-
 Even $[2n]$ form $\rightarrow$ 0 is even no.
 odd $[2n+1]$ form

Imp :-

odd $\times$ odd = odd

Even $\times$ Even = Even

odd $\times$ Even = Even

odd $\div$ odd = Even

even $\div$ even = even

odd $\div$ even = odd

PB :-

$$\begin{aligned} x+y &= A & \rightarrow & x = \frac{A+B}{2} \\ x-y &= B & \rightarrow & y = \frac{A-B}{2} \end{aligned}$$

perfect sq. $\rightarrow n^2$

perfect cube $\rightarrow n^3$

perfect nos $\rightarrow$ sum of proper divisor (excluding itself)
 eg $\rightarrow$ 6 $\rightarrow$ 1+2+3 = 6
 28 $\rightarrow$ 1+2+4+7+14 = 28

① face value - 671234

face value of 7 = 7.

② place value - 67324.

place value of 7 = 7000.

REMAINDER THEOREM

Dividend
 Divisor - 65
 Quotient
 Remainder

Dividend = Divisor $\times$ Quotient + Rem.

Ex: - If, divisor is D then remainder can be 0 to (D-1).

- 132 is completely divisible by 22.
- 22 is a factor of 132.
- 132 is a multiple of 22.

If we divide greater no. with smaller no. then divisor will be greater than remainder.

eg - $\frac{15}{7} \rightarrow 1 \text{ rem}$

If we divide smaller no. with greater no. then rem will be the same smaller no.

eg. $\frac{9}{15} \rightarrow 0 \text{ rem}$

Ex 2.

Factor
 not factor
 G.N.D.

part method NO.
 $\frac{N}{7} = \frac{218 + 11}{7} = 31 \text{ rem } 4$

Ex 3
 $7 \div 288 \rightarrow 83 \text{ rem.}$
 $7 \div 15 \rightarrow 2 \text{ (rem.)}$
 $2 = ?$

soln $\rightarrow \frac{7}{15} \rightarrow 83 \text{ rem.} \mid \frac{7}{15} \rightarrow 83 \rightarrow 3 \text{ (rem.)}$

Ex 4
 $\frac{N}{16} 211 \text{ (form.)}$
 $\frac{507}{16} 2 \frac{5}{16} = \frac{48+7}{16} \text{ (2)}$

Ex 5: a, b are +ve integers
 $a^n \pm b^n \rightarrow \begin{cases} \div (a+b) \\ \div (a-b) \end{cases}$

n odd.
 $a^n + b^n \div a+b \checkmark$
 $a^n - b^n \div a-b \checkmark$

n even.
 $a^n + b^n \div a+b \times$
 $a^n - b^n \div a-b \checkmark$

n even.
 $a^n + b^n \div a+b \times$
 $a^n - b^n \div a-b \checkmark$

n even.
 $a^n + b^n \div a+b \checkmark$
 $a^n - b^n \div a-b \checkmark$

$a^n + b^n$ is a factor of $(a+b)$.

eg $\rightarrow \frac{2735 + 115}{(2+11)} = 0 \text{ (rem.)}$
 $\because 38 = 2 \times 19$

DIVISIBILITY RULES

1 is not divisible by any No. except 1 but 1 is a universal factor.

2, 4, 8, 16, 32 & 64 ka divisibility

rule:-

2 → last two digit should be even or, last digit should be ÷ by 2.
eg → 1328

4 → last two digits should be ÷ by 4
eg → 756

8 → last three digits ÷ by 8
eg → 34584

16 → last 4 digits ÷ by 16
eg → 6846512

5, 25, 125 ka divisibility rule:-

5 → last 1 digit ÷ by 5

25 → last 2 digit ÷ by 5

125 → last 3 digit ÷ by 5

7, 11, 13 ka divisibility rule:-

003542
diff = 835
7x ✓
11 ✓
13 x

683224
diff = 429
7x ✓
11 ✓
13 ✓

002973402432
diff = 1405404 = 100
7x ✓
11 ✓
13 ✓

7x11x13 = 1001
special no.

Special 11 ka divisibility rule:-

- Add even place digits
- Add odd place digits

eg → 166452

16 + 5 = 21
6 + 4 + 2 = 12
diff = 9
if diff = 0 or multiple of 11 then no. ÷ by 11

3, 9 ka divisibility rule

3 → sum of digits ÷ by 3

9 → sum of digits ÷ by 9

Digital sum:- किसी भी सं. को 9 से भाग देने पर जो remainder आता है, उसे उस सं. का digital sum कहा जाता है।

Digital sum is a single digit No. always smaller than 9.

eg 72 → 0/9

4567 X 615 = 2808705

4 X 3 2

3 → 3

10s 10s

→ should become

D.S. → the D.S. of a no. means the sum of all its digits.

eg 4567 X 615 = 2808705
D.S. = 4+5+6+7+6+1+5 = 34
34 ÷ 9 = 3 remainder 7
7 ÷ 9 = 0 remainder 7
D.S. = 7

N / 5x2 → 1 | N / 4x1 → 1 | N / 8x8 → 1

N / 3, 6, 21, 9 → ?

Rem of 11?

54813702
- 1 7 4 2

-17 + 422 = 405

aaa2ax11 = ax3x37

abab : abx101

abcabc2 : abcx1001 = abcx7x13

ababab2 : abx10101 = abx3x7x13

Negative Remainder Concept

$\rightarrow 79 \div 13 \rightarrow 5$ or $-5 + 13 = 8$
 $\rightarrow 3 \times 2 \times 1 \times 5 = 30$ or $-30 + 59 = 29$
 $\rightarrow 56 \times 9 \times 58 \times 64 = 200$ or $-30 + 59 = 29$

Factorial

$1! = 1$
 $2! = 2$
 $3! = 6$
 $4! = 24$
 $5! = 120$
 $6! = 720$
 $7! = 5040$

$$n! = n(n-1)(n-2) \dots 1$$

$$n! = n \times (n-1)!$$

$$0! = 1$$

Fermat's Theorem

$$\frac{a^{p-1}}{p} = 1 \text{ (rem)}$$

where, p = prime No.
 $a, p \in \text{Co-prime}$

eg $\rightarrow \frac{2^{101}}{101} = \frac{2^{100} \times 2}{101} = 1 \times 2 = 2 \text{ (rem)}$

Theorem

$\frac{(a+k)^n}{a} = 1 \text{ (rem)}$
 $\frac{(a-k)^n}{a} = (-1)^n \text{ (rem)}$
 eg $\rightarrow \frac{1686}{15} = 1$
 $\frac{1437}{18} = -1$ or 17

Special case: $\frac{4^n}{6} = 1 \text{ (rem)}$

Euler's Theorem

$$\frac{a^{\phi(N)}}{N} = 1 \text{ (rem)}$$

N = Natural No.

$\phi(N)$ = Totient function of N

$a, N \in \text{Co-prime}$

meaning of $\phi(N)$:

$\phi(N)$ = N se dote ki kintni se N ke saath co-prime hai.

eg $N = 12 = 2^2 \times 3$
 $\phi(N) = 12 \times \frac{1}{2} \times \frac{2}{3} = 4$

$12 \Rightarrow (1, 3, 5, 7, 11)$

Co-prime $\rightarrow (12, 1) (12, 5) (12, 7) (12, 11)$

$\rightarrow 12$ se dote ki 4 nos. 12 ke saath co-prime hai.

eg: $\frac{357}{10} = 7 \text{ (rem)}$

$\frac{5489}{100} = 8 \text{ (rem)}$

$\frac{7386124}{1000} = 124 \text{ (rem)}$

$\rightarrow \frac{999999}{1000} = 999 \times 1000 = 999000$
 $= 9 \times 3 \times 3 \times 7 \times 11 \times 13$
 eg $\frac{888888}{39} = 0 \text{ (rem)}$
 $\downarrow$
 13×3

eg $\rightarrow$ The largest No. which always divides product of any n consecutive natural no's $= n!$

$n(n+1)(n+2)(n+3)(n+4)$ is always divisible by $5! = 120$

Consecutive Rem

$(12) 987 \div 823$
 96
 27
 24
 34
 36
 1

$D \rightarrow \text{HCF}[56, 24, 36]$
 $= 12$

consecutive rem $\rightarrow 2, 3, 1$

Successive Division

→ 796 को 5, 7 & 4 से suc. divide

$$\begin{array}{r} 5 \overline{) 796} \quad 7 \overline{) 159} \quad 4 \overline{) 22} \\ \underline{-155} \quad \underline{-154} \quad \underline{-20} \\ 1 \quad 5 \quad 2 \end{array}$$

successive rem → 1, 5, 2

$$\begin{array}{r} 5 \overline{) 796} \times 1 \\ \underline{795} \quad 5 \\ 7 \overline{) 159} \times 5 \\ \underline{354} \quad 5 \\ 4 \overline{) 22} \times 2 \\ \underline{22} \quad 2 \\ 1 \end{array}$$

↓ final quotient

→ find the least no. which is suc. by 5, 7, 7 & 4 leaves rem → 1, 5, 2?

∴ standard form:-

$$N = (\text{Prod. of } 5, 7, 4) \cdot K + 96$$

$$N = 140K + 96$$

$$N = 140 \times 1 + 96 \quad (K=1)$$

$$= 236 \rightarrow \text{smallest no.}$$

UNIT DIGIT

digits (0-9):

$$\begin{array}{ll} (-0)^2 \equiv 0 & (-5)^2 \equiv 5 \\ (-1)^2 \equiv 1 & (-6)^2 \equiv 6 \\ (-2)^2 \equiv 4 & (-7)^2 \equiv 9 \\ (-3)^2 \equiv 9 & (-8)^2 \equiv 4 \\ (-4)^2 \equiv 6 & (-9)^2 \equiv 1 \end{array}$$

least perfect sq.
को unit digit
②, ③, ⑦ & ⑧
को ही हो पाता है.

→ 5X even no → 0
5X odd no → 5
→ 0, 1, 5, 6 → any power unit digit
will be same

$$\rightarrow 4 \rightarrow (-4)^{\text{odd}} \equiv 4 \quad (-4)^{\text{even}} \equiv 6$$

→ 2nd power को

$$\begin{array}{l} 2^1 \equiv 2 \\ 2^2 \equiv 4 \\ 2^3 \equiv 8 \\ 2^4 \equiv 6 \\ 2^5 \equiv 2 \\ 2^6 \equiv 4 \\ 2^7 \equiv 8 \\ 2^8 \equiv 6 \\ 2^9 \equiv 2 \\ 2^{10} \equiv 4 \end{array}$$

→ 4 power के बाद unit digit repeat होती है।
②, ③, ⑦ & ⑧ के case में यही होता है

$$\text{Power } \div 4$$

$$\begin{array}{l} 1 \rightarrow 2^1 \equiv 2 \\ 2 \rightarrow 2^2 \equiv 4 \\ 3 \rightarrow 2^3 \equiv 8 \\ 0 \rightarrow 2^4 \equiv 6 \end{array}$$

→ unit digit of (2^n) unit digit of (2^{n+4})

eg if $472^n \rightarrow \text{u.d.} \equiv 8$
then, $472^{n+36} \rightarrow \text{u.d.} \equiv 8$

AP, SP & HP

ARITHMETIC PROGRESSION (A.P.):

→ nth term of AP: also the last term

$$T_n = a + (n-1)d$$

last term

$$\text{No. of terms } (n) = \frac{\text{last term} - \text{first term}}{d} + 1$$

common diff

→ sum of n terms:-

$$S_n = \frac{n}{2} (a + l)$$

$$\text{OR} \quad S_n = \frac{n}{2} [2a + (n-1)d]$$

eg → 5, 8, 11, 14, 17

$$\text{Avg} = \frac{5+17}{2} = 11$$

$$\text{Avg} = \frac{14+18}{2} = 11$$

$$1+15+2+14$$

Avg 2 middle no. = 11

$$S_n = \left(\frac{5+17}{2} \right) \times 5$$

$$S_n = \text{avg} \times n$$

$$2 \times \frac{a+l}{2} \times n = \frac{n}{2} (a+l)$$

① Sum of first n natural Nos.

$$1+2+3+\dots+n = \sum_{n=1}^n n = \frac{n(n+1)}{2}$$

② Sum of first n even natural Nos.:

$$2+4+6+\dots+2n = \sum_{n=1}^n 2n = n(n+1)$$

③ Sum of first n odd natural Nos.:

$$1+3+5+\dots+(2n-1) = \sum_{n=1}^n (2n-1) = n^2$$

④ Sum of squares of first n natural Nos.

$$1^2+2^2+3^2+\dots+n^2 = \sum_{n=1}^n n^2 = \frac{n(n+1)(2n+1)}{6}$$

⑤ Sum of cubes of first n natural Nos.

$$1^3+2^3+3^3+\dots+n^3 = \sum_{n=1}^n n^3 = \left[\frac{n(n+1)}{2} \right]^2$$

⑥ Sum of 4th power of first n natural Nos.:

$$1^4+2^4+3^4+\dots+n^4 = \sum_{n=1}^n n^4 = \frac{n(n+1)(6n^3+9n^2+n-1)}{30}$$

GEOMETRIC PROGRESSION (G.P.)

Standard form: $a, ar, ar^2, ar^3, \dots, ar^{n-1}$

$\rightarrow$ n^{th} term of a G.P. $(T_n) = a \cdot r^{n-1}$

$\rightarrow$ Sum of n terms of a G.P. $S_n = a \frac{r^n - 1}{r - 1}$

Sum of infinite terms of a G.P. series

if $|r| > 1$

$$S_{\infty} = \infty$$

$$2+6+18+54+\dots = \infty$$

$$|r| > 1$$

if $|r| < 1$

$$S_{\infty} = \frac{a}{1-r}$$

$$1 + \frac{1}{6} + \frac{1}{36} + \dots = \infty = \frac{1}{1-\frac{1}{6}}$$

$$= \frac{6}{5}$$

$$64 + 48 + 36 + 24 + 16 + \dots = \infty$$

$$= \frac{64}{1-\frac{3}{4}} = 256$$

Harmonic progression (H.P.)

$$A.P. \rightarrow 3, 10, 13, 16 \dots$$

$$H.P. \rightarrow \frac{1}{3}, \frac{1}{10}, \frac{1}{13}, \frac{1}{16} \dots$$

if a & b are two terms then,

$$A.M. = \frac{a+b}{2}$$

$$G.M. = \sqrt{ab}$$

$$H.M. = \frac{2ab}{a+b}$$

$$A.M. \geq G.M. \geq H.M.$$

$$A.M. \times H.M. = G.M.^2$$

$\rightarrow$ if a, b & c are in A.P.

$$b-a = c-b \Rightarrow 2b = a+c$$

$$b = \frac{a+c}{2}$$

$\rightarrow$ if a, b & c are in G.P.

$$\frac{b}{a} = \frac{c}{b} \Rightarrow b^2 = ac$$

$$b = \sqrt{ac}$$

$\rightarrow$ if $a_1, a_2, a_3, a_4, \dots, a_n$ are n terms in H.P.

$$\frac{a_1+a_2+a_3+\dots+a_n}{n} \geq \sqrt[n]{a_1 a_2 a_3 \dots a_n} \geq \frac{1}{\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n}}$$

NUMBER OF FACTORS

$\rightarrow$ How to find factors of a No.?

$$\text{eg } 72 \rightarrow 2^3 \times 3^2$$

$$T.N.F. = (3+1) \times (2+1) = 12$$

$$2^0, 2^1, 2^2, 2^3$$

$$3^0, 3^1, 3^2$$

$$1, 2, 3, 4, 6, 8, 9, 12, 16, 18, 24, 27, 36, 48, 54, 72$$

$\rightarrow$ Sum of factors:

$$72 \rightarrow 2^3 \times 3^2$$

$$[2^0+2^1+2^2+2^3] \times [3^0+3^1+3^2]$$

$$= 15 \times 13 = 195$$

Ex $2160 = 2^4 \times 3^3 \times 5^1$

$TNF = (4+1)(3+1)(1+1) = 40$

→ Sum of factors:

$[2^0 + 2^1 + 2^2 + 2^3 + 2^4] \times [3^0 + 3^1 + 3^2 + 3^3] \times [5^0 + 5^1]$

$\Rightarrow 31 \times 40 \times 6 = 7440$

$2160 = 2^4 \times 3^3 \times 5^1$

No. of odd factors $= (3+1) \times (1+1) = 8$

Sum of odd factors $= [3^0 + 3^1 + 3^2 + 3^3] \times [5^0 + 5^1] = 240$

No. of even factors $= 2160 = 2 \times [2^3 \times 3^3 \times 5^1]$

Sum of even factors $= [2^1 + 2^2 + 2^3 + 2^4] \times [3^0 + 3^1 + 3^2 + 3^3] \times [5^0 + 5^1]$

$= 7200$

$2160 = 2^4 \times 3^3 \times 5^1 = 2^2 \times 3 \times [2^2 \times 3^2 \times 5^1]$

① No. of factors which are multiple of 12:-
 $3 \times 3 \times 2 = 18$

② Sum of factors which are multiple of 12:-

$[2^2 + 2^3 + 2^4] \times [3^1 + 3^2 + 3^3] \times [5^0 + 5^1]$

$= 28 \times 39 \times 6 = 6552$

→ Sum of reciprocals of factors of a Nos:-

$10 \rightarrow 1, 2, 5, 10$

$\frac{1}{1} + \frac{1}{2} + \frac{1}{5} + \frac{1}{10} = \frac{10+5+2+1}{10} = \frac{18}{10} = \frac{9}{5}$

$N = 2^4 \times 3^3 \times 5^1$
→ No. of perfect sq. factors $= [2^0 2^2 2^4 2^6 2^8 2^{10}] \times$

$[3^0 3^2 3^4 3^6] \times [5^0 5^2 5^4]$

Trick $\Rightarrow (5+1) \times (3+1) \times (2+1) = 72$

→ No. of perfect cube factors:-

$(3+1) \times (2+1) \times (1+1) = 24$

Notes N even $\rightarrow$ perfect sq.

N 3rd multiple $\rightarrow$ perfect cube

→ Both perfect sq. & perfect cube factors of N:-

$N = 2^{20} \times 3^5 \times 5^1 \times 7^7$

$= (3+1) \times (2+1) \times (1+1) \times (1+1) = 48$

→ If given Nos is not a perfect sq.

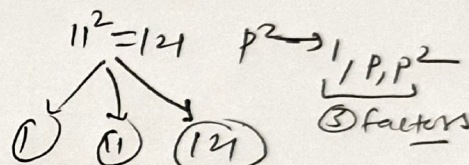
→ Even No. of factors.

→ If given Nos is a perfect sq.

→ Odd no of factors

→ No. of factors of product $= \frac{T \cdot n f}{2}$

② sq. of any prime no. has exactly 3 factors:-



END